STUDYHUB
CLASS 12 — MP BOARD
MATHEMATICS — SET 2
ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 80
Medium: English	Set: 2

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 contain objective-type sub-questions.
3. Questions 6 to 15 carry 2 marks each.
4. Questions 16 to 19 carry 3 marks each.
5. Questions 20 to 23 carry 4 marks each.
6. Internal choices are given wherever indicated.
7. Use of calculator is not permitted.

SECTION – A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose and write the correct option. (1×6=6)

(i) If $f(x)=1/x$, then f is:
A. One-one but not onto
B. Onto but not one-one
C. One-one and onto
D. Neither one-one nor onto

(ii) The value of $\tan^{-1}(1)+\tan^{-1}(1)$ is:
A. $\pi/4$
B. $\pi/2$
C. π
D. 2π

(iii) If $A=[[1,0],[0,1]]$, then A^2 is:
A. Zero matrix
B. A
C. 2A
D. A^{-1}

(iv) If $y=\log x$, then dy/dx is:
A. x
B. $1/x$
C. $\log x$
D. e^x

(v) The scalar projection of $\vec{a}$ on $\vec{b}$ is:
A. $\vec{a}\times\vec{b}$
B. $(\vec{a}\cdot\vec{b})/|\vec{b}|$
C. $|\vec{a}||\vec{b}|$
D. $\vec{a}+\vec{b}$

(vi) If $P(A)=0.4$, then $P(A')$ is:
A. 0.2
B. 0.4
C. 0.6
D. 1.4

Q.2 Fill in the blanks. (1×6=6)

(i) The inverse of a bijective function always ______.
(ii) $\cos^{-1}(1)=$ ______.
(iii) If A is a square matrix, then $AA^{-1}=$ ______.
(iv) $d/dx(x^n)=$ ______.
(v) The cross product of two parallel vectors is ______.
(vi) The probability of a certain event is ______.

Q.3 Write True or False. (1×6=6)

(i) Every bijective function has an inverse.
(ii) $\sin^{-1}x$ is defined for all real values of x.
(iii) The determinant of a triangular matrix is equal to the product of its diagonal elements.

(iv) A differentiable function is always continuous.
(v) The scalar product of two vectors is always a vector.
(vi) The sum of probabilities of all elementary outcomes of a random experiment is 1.

Q.4 Match Column A with Column B. (1×7=7)

Column A	Column B
(i) $\int e^x\,dx$	(a) $1/(1+x^2)$
(ii) $d/dx(\sin^{-1}x)$	(b) e^x+C
(iii) $d/dx(\cos x)$	(c) $-\sin x$
(iv) $\int dx/(1+x^2)$	(d) $\tan^{-1}x+C$
(v) $\lvert\vec{a}\times\vec{b}\rvert$	(e) $\lvert a\rvert\lvert b\rvert\sin\theta$
(vi) $P(A')$	(f) $1-P(A)$
(vii) $d/dx(x^n)$	(g) $nx^{n-1})$

Q.5 Answer in one word/one sentence. (1×7=7)

(i) What type of function has both one-one and onto properties?
(ii) Write the principal value of $\sin^{-1}(1)$.
(iii) What is the determinant of an identity matrix of order 3?
(iv) Find the derivative of cos x.
(v) What is the angle between two parallel vectors having the same direction?
(vi) Write the order of $dy/dx+y=x$.
(vii) If $P(E)=1$, what type of event is E.

SECTION – B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)
Let $f:R\to R$, $f(x)=x^2$. Is f one-one? Give reason.
OR
Find the inverse of $f(x)=3x-5$.

Q.7 (2 Marks)
Prove that $\sin^{-1}x+\cos^{-1}x=\pi/2$.
OR
Evaluate $\tan^{-1}(1/2)+\tan^{-1}(1/3)$.

Q.8 (2 Marks)
If A=[[2,3],[1,4]], find the adjoint of A.
OR
If |[[x,2],[3,4]]|=10, find x.

Q.9 (2 Marks)
Differentiate $y=x^2\sin x$.
OR
Find dy/dx, if $y=(x+1)/(x-1)$.

Q.10 (2 Marks)
Find the critical points of $f(x)=x^3-3x^2+2$.
OR
Find the minimum value of $x+4/x$, $x>0$.

Q.11 (2 Marks)
Evaluate $\int x\cos x\,dx$.
OR
Evaluate $\int dx/\sqrt{(1-x^2)}$.

Q.12 (2 Marks)
Evaluate $\int_0^\pi \sin x\,dx$.
OR
Find the area under $y=x^2$ from $x=0$ to $x=1$.

Q.13 (2 Marks)
Solve $dy/dx=y/x$.
OR
Solve $dy/dx+3y=0$.

Q.14 (2 Marks)
Find the projection of $\vec{a}=2i+3j+6k$ on $\vec{b}=i+2j+2k$.
OR
Find a vector perpendicular to both $i+j$ and $i-j$.

Q.15 (2 Marks)
A coin is tossed three times. Find the probability of exactly two heads.
OR
Two dice are thrown simultaneously. Find the probability of getting a sum of 7.

SECTION – C : SHORT ANSWER QUESTIONS

Q.16 (3 Marks)
Using matrices, solve $x+2y=8$, $3x-y=3$.
OR
Solve by matrix method: $2x-y=4$, $x+3y=9$.

Q.17 (3 Marks)
Find the intervals in which $f(x)=x^3-3x^2-9x+5$ is increasing or decreasing.
OR
Find the absolute maximum and minimum values of $f(x)=x^2-6x+8$ on $[1,5]$.

Q.18 (3 Marks)
Evaluate $\int(x+1)/(x^2+2x+5)\,dx$.
OR
Evaluate $\int x^2e^x\,dx$.

Q.19 (3 Marks)
Find the equation of the plane passing through (1,2,3), (2,−1,1), (3,0,2).
OR
Find the direction cosines of the line joining A(1,2,3) and B(4,6,3).

SECTION – D : LONG ANSWER QUESTIONS

Q.20 (4 Marks)
Using integration, find the area enclosed between $y=x+2$ and $y=x^2$.
OR
Evaluate $\int_0^1 x/(1+x^2)\,dx$.

Q.21 (4 Marks)
Solve the differential equation $dy/dx+2xy=2x$.
OR
Solve $dy/dx+2y/x=x^2$.

Q.22 (4 Marks)
A bag contains 6 white and 4 black balls. Three balls are drawn at random without replacement. Find the probability that (i) all three are white, (ii) exactly two are black.
OR
From a pack of 52 playing cards, two cards are drawn without replacement. Find the probability that (i) both are aces, (ii) one is an ace and the other is a king.

Q.23 (4 Marks)
Find the shortest distance between $r=(2i-j+k)+\lambda(i+2j-k)$ and $r=(i+3j-2k)+\mu(2i-j+2k)$.
OR
Find the equation of the plane passing through (2,−1,4) and parallel to $i+2j-k$ and $2i-j+3k$.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	6
Q.4	7
Q.5	7
Q.6–Q.15	20
Q.16–Q.19	12
Q.20–Q.23	16
TOTAL	80

6 + 6 + 6 + 7 + 7 + 20 + 12 + 16 = 80 Marks

SHORT ANSWER KEY

Q.1: C, B, B, B, B, C. Q.2: exists; 0; I; nx^{n-1}; zero vector; 1. Q.3: True, False, True, True, False, True. Q.4: (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d), (v)-(e), (vi)-(f), (vii)-(g). Q.5: Bijective function; $\pi/2$; 1; $-\sin x$; 0°; 1; Certain event.

For Q.6–Q.23, evaluate according to correct mathematical method, steps, formulae and final answer. Full solutions are intentionally not included.